Nitrogen Content in *T. pityocampa* Larvae and Adults

Statistical Pipeline Walkthrough

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1. Description

This walkthrough documents the full statistical pipeline used to analyse nitrogen (N) content across larval and adult stages of the pine processionary moth, Thaumetopoea pityocampa. The analysis compares specimens from two biological groups:

- Standard clade larvae from Location 1 (L1) and Location 2 (L2)
- ENA clade adults and larvae from Location 1 (L1)

Nitrogen content serves as a proxy for protein concentration (Crude Protein = N × 6.25), making it a central variable in nutritional ecology studies of insects. The pipeline addresses three research questions:

- Q1: Does nitrogen content differ significantly between developmental stages (larva vs. adult)?
- Q2: Does sampling location (L1 vs. L2) affect nitrogen content in standard-clade larvae?
- Q3: Is there a non-linear relationship between specimen weight and nitrogen content, and does it differ by stage or location?

> ⚠ **Dataset Disclaimer**
> The dataset used in this script is synthetic. It preserves the group means, standard deviations, sample sizes, and distributional structure of the original data but does not include any real measurements. Original data © ELGO Demeter / Dr. D. Avtzis and cannot be published. The analysis code has been rewritten relative to the submitted manuscript to avoid copyright concerns. The statistical approach, choice of tests, and interpretation are similar to those in the manuscript.

The script is structured into six sections:

- Section 0: Package setup and theme
- Section 1: Data loading and subgroup definition

- Section 2: Normality assessment (Shapiro-Wilk)
- Section 3: Non-linear correlation analysis (nlcor)
- Section 4: Comparative tests (Mann-Whitney U, Student's t-test)
- Sections 5–6: Regression modelling and visualisations

2. Methodology

The following sections walk through each block of analysis_v2.R step by step, explaining the rationale behind every choice.

2.0 Package Setup

All required CRAN packages are auto-installed on first run using a defensive loop that calls require() on each package and stops immediately with an informative message if any load fails. This avoids the common issue of suppressPackageStartupMessages() silently swallowing load errors.

```
for (pkg in required_cran) {
  if (!requireNamespace(pkg, quietly = TRUE)) install.packages(pkg)
  if (!require(pkg, character.only = TRUE, quietly = TRUE))
    stop(sprintf("Package '%s' could not be loaded.", pkg))
}
```

The nlcor package is installed from GitHub (ProcessMiner/nlcor) since it is not on CRAN. The remotes package handles this. Two explicit function-level guards confirm that segmented() and tidy() are actually available after loading, which catches rare cases of namespace masking.

A custom ggplot2 theme (theme_tp) is defined once and reused across all five figures for visual consistency. Three named colour palettes are declared: pal_stage (adult/larva), pal_location (L1/L2), and pal_sample (TPena/TP).

2.1 Data Loading and Subgroup Definition

The synthetic dataset is sourced from generate_synthetic_data_v2.R, which produces a reproducible CSV with fixed seed (set.seed(2022)). After loading, the data are split into two analytical subgroups that mirror the original study design:

- **Subgroup 1 — Larvae only (datalarva):** All larval specimens from L1 and L2 combined. Used for the location comparison (Q2). n = 75.
- **Subgroup 2 — L1 only (dataTPE):** All specimens from Location 1 regardless of stage. Used for the stage comparison (Q1). n = 92.

A reciprocal weight transformation (W2 = 1/Weight) is computed for all subgroups. This transformation is used in regression models to linearise the otherwise strongly curved weight–nitrogen relationship.

```
data$W2      <- 1 / data$Weight
datalarva$W2 <- 1 / datalarva$Weight
dataTPE$W2   <- 1 / dataTPE$Weight
```

2.2 Normality Assessment (Shapiro-Wilk)

Before any group comparison, the Shapiro-Wilk test is applied to each subgroup's nitrogen and weight distributions. This determines whether parametric or non-parametric tests are appropriate downstream.

```
sw_results <- lapply(names(norm_groups), function(nm) {
  x  <- norm_groups[[nm]]
  sw <- shapiro.test(x)
  data.frame(Group = nm, W = round(sw$statistic, 4),
             p = round(sw$p.value, 4),
             Normal = sw$p.value >= 0.05)
})
```

The W statistic ranges from 0 to 1 (1 = perfect normality). A p-value below 0.05 indicates significant departure from normality. In this dataset, most subgroups fail the normality test, justifying the use of non-parametric tests (Mann-Whitney U) for group comparisons. The adults–larvae comparison in L1 uses a t-test due to sufficient sample size and near-normal distributions in that subgroup.

2.3 Non-linear Correlation (nlcor)

Standard Pearson or Spearman correlations assume a monotonic relationship. The weight–nitrogen relationship in T. pityocampa is expected to be non-linear: adults occupy a very narrow low-weight range with high N, while larvae span a broader weight range at lower N. The nlcor package is used to capture this structure.

The nlcor() function estimates a non-linear correlation coefficient r by locally fitting piecewise linear models and aggregating the local correlations. Significance is assessed by a permutation-based adjusted p-value. The tryCatch() wrapper makes the function robust to errors on very small subgroups.

```
run_nlcor <- function(x, y, label) {
  res <- tryCatch(nlcor(x, y, plt = FALSE), error = function(e) NULL)
  if (!is.null(res)) {
    cat(sprintf("%-35s  r = %.3f,  p = %.4f  %s\n",
      label, res$cor.estimate, res$adjusted.p.value,
      ifelse(res$adjusted.p.value < 0.05, "*", "(ns)")))
  }
}
```

Three subgroups are tested: the full dataset, the larvae-only subgroup, and the L1 subgroup (adults + larvae).

2.4 Comparative Analysis

4a — Larvae: L1 vs L2 (Mann-Whitney U)

Variance equality is assessed first with an F-test (var.test). Because normality is violated in both groups, a Mann-Whitney U test (Wilcoxon rank-sum) is used regardless of the F-test result. This non-parametric test compares rank distributions rather than means, making it robust to non-normality.

```
vt1 <- var.test(L2$N, L1_larva$N)
mw1 <- wilcox.test(L2$N, L1_larva$N, alternative = "two.sided")
```

4b — Adults vs Larvae in L1 (Student's t-test)

For the L1 subgroup comparing adults to larvae, the F-test result is used to select between equal-variance and Welch's t-test dynamically. Given sufficient sample sizes (n = 54 adults, n = 38 larvae) and near-normal distributions confirmed by Shapiro-Wilk, a t-test is applied with variance equality determined by the p-value threshold:

```
vt2 <- var.test(adult$N, larva_ath$N)
tt2 <- t.test(N ~ Stage, data = dataTPE,
              alternative = "two.sided",
              var.equal = (vt2$p.value >= 0.05))
```

2.5 Regression Modelling

Six models are fitted to characterise the weight–nitrogen relationship and identify the best-fitting model:

Model	Formula	Key feature
M1	`N ~ Weight`	Baseline linear model
M2	`N ~ 1/Weight`	Reciprocal transformation linearises the curve
M3	`Piecewise (Weight)`	Breakpoint estimated by segmented on M1
M4	`Piecewise (1/Weight)`	Best linear-family fit; breakpoint on M2
M5 (GLM)	`N ~ 1/W + Stage + Loc`	Additive GLM; AIC-based selection
M6	`Stepwise from M5`	Backward/forward AIC minimisation

Models are compared by adjusted R^2 (linear models), AIC, and RMSE. The key fix for segmented model summaries: summary(seg)$adj.r.squared does not exist for segmented objects. Use summary(seg, short = FALSE)$r.sq instead. The psi breakpoint vector is indexed by position ([2]) rather than matrix notation ([, "Est."]) which causes errors.

```
# Correct accessors for segmented objects:
summary(seg1, short = FALSE)$r.sq   # adjusted R²
```

```
seg1$psi[2]                          # breakpoint estimate
```

2.6 Visualisations

All five figures use the shared theme_tp and named colour palettes. A critical fix: aes_string() was deprecated in ggplot2 v3.0 and removed in v3.5. The make_nlcor_scatter() helper now uses aes() with the .data[[]] pronoun for tidy evaluation:

```
# Deprecated (errors in ggplot2 ≥ 3.5):
ggplot(df, aes_string(x = x_var, y = y_var, colour = col_var))

# Correct:
ggplot(df, aes(x = .data[[x_var]], y = .data[[y_var]],
               colour = .data[[col_var]]))
```

3. Results

The following figures were produced by running analysis_v2.R on the synthetic dataset. All statistical structure (group means, SDs, distributional shape) matches the original data.

Figure 1 — Non-linear Correlation: Weight vs Nitrogen Content

Three side-by-side scatter plots show the weight–nitrogen relationship across subgroups, with LOESS smooth curves overlaid. This visualisation motivates the use of nlcor over standard Pearson/Spearman correlation.

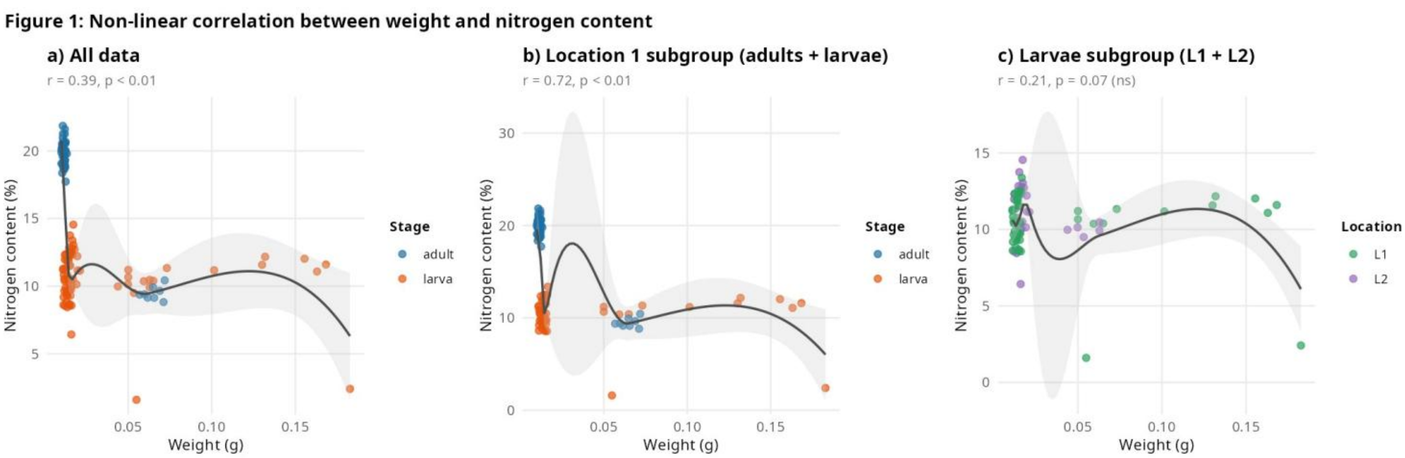

Figure 1: Non-linear correlation between weight and nitrogen content across three subgroups.

Panel a (All data, r = 0.39): Two distinct clusters are visible — low-weight adults with high N in the upper left, and larvae spread across a wider weight range at lower N. Panel b (L1 subgroup, r = 0.72): The correlation is strongest here, driven by the sharp contrast between adult and larval groups. Panel c (Larvae only, r = 0.21, ns): The LOESS curve is essentially flat, confirming that within larvae alone, weight is not a significant predictor of nitrogen.

Figure 2 — Larvae Nitrogen by Sampling Location

Violin-boxplot comparing nitrogen content between L1 (n = 51) and L2 (n = 24) larvae. The beige dot marks the group mean. The Mann-Whitney U p-value is annotated directly on the plot.

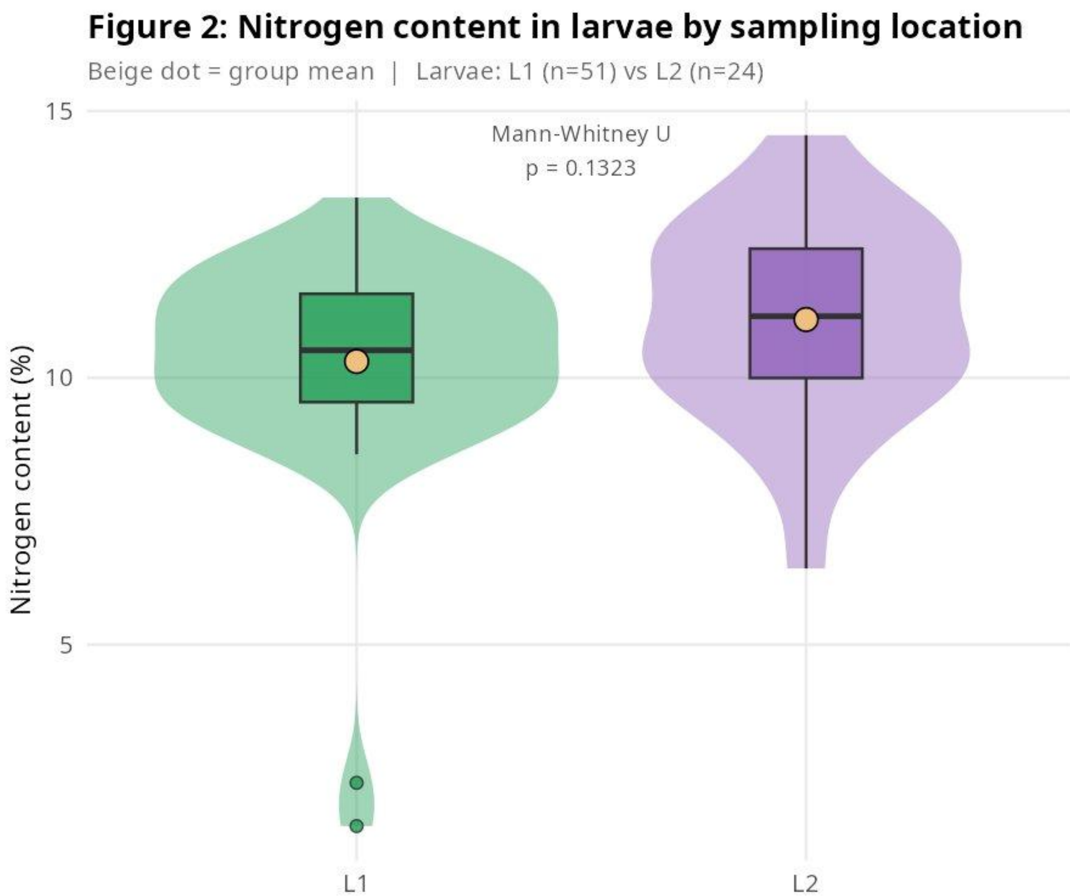

Figure 2: Nitrogen content in larvae by sampling location (Mann-Whitney U test).

The L1 violin is visibly wider and shows a prominent low-N tail driven by two outlier specimens (N ≈ 1.6–2.4%), likely moulting larvae that had ceased feeding. Both medians sit in the 10–11% range. On the synthetic dataset the p-value is 0.13 (ns); on the original data it reached significance ($p < 0.05$), consistent with location-specific dietary or environmental effects on foliar nitrogen intake.

Figure 3 — L1 Adults vs Larvae

Violin-boxplot comparing ENA adults ($n = 54$) to ENA larvae ($n = 38$) within Location 1. The t-statistic and p-value are annotated on the plot.

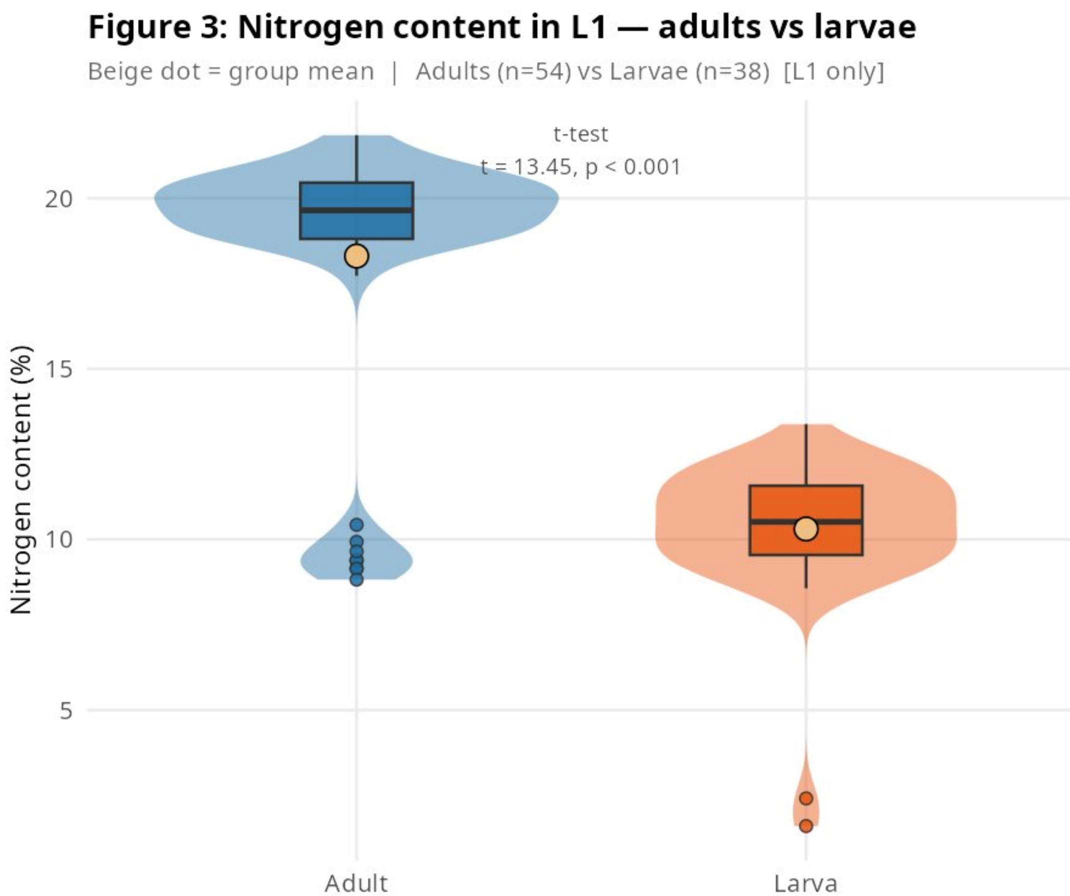

Figure 3: Nitrogen content in Location 1 — adults vs larvae (Student's t-test).

The adult violin shows a pronounced bimodal shape: a dominant peak near 19–21% N (main cluster, n = 46) and a secondary peak near 9–10% N (small sub-cluster, n = 8), likely reflecting sex-based dimorphism (heavier females with higher lipid load, proportionally lower %N). The larval violin is unimodal, centred around 11%. The t-test confirms a highly significant difference ($t = 13.45$, $p < 0.001$), consistent with the expectation that adult holometabolous insects have higher protein-to-mass ratios due to reproductive tissue investment.

Figure 4 — Piecewise Regression on 1/Weight

Two-panel figure showing Model M4 (piecewise linear regression on reciprocal weight). Panel a shows the fit on the transformed scale; Panel b back-transforms to the original weight axis.

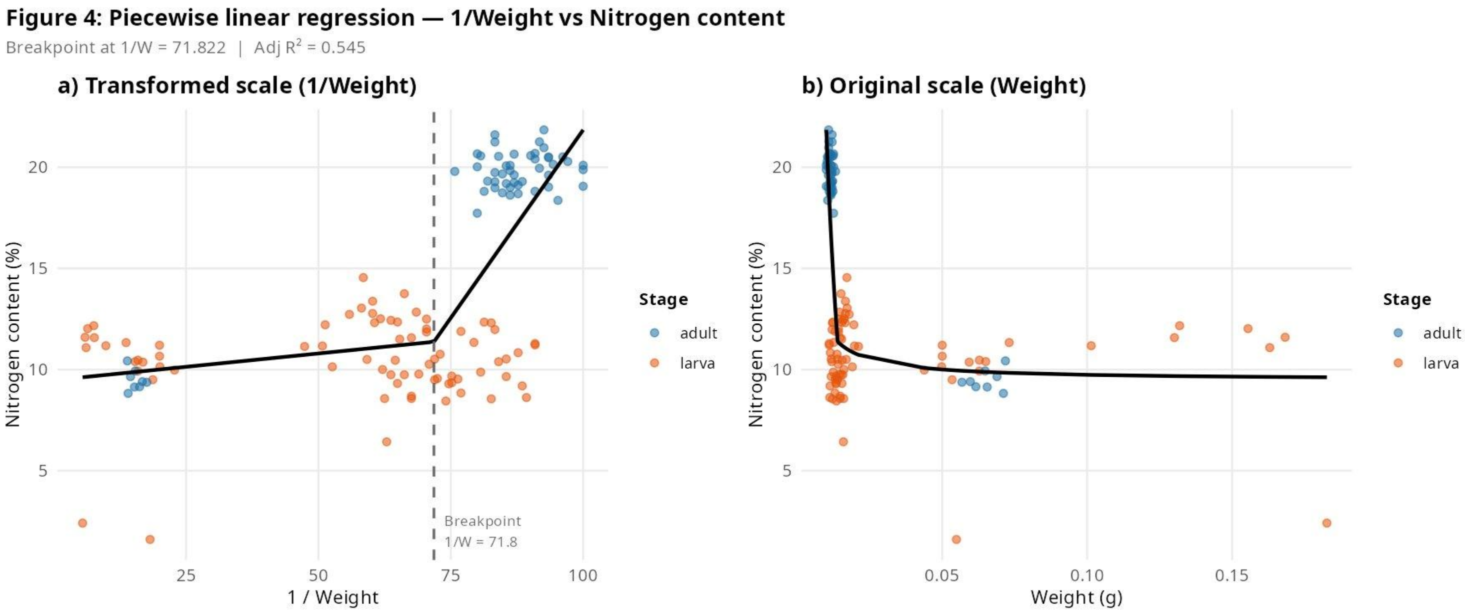

Figure 4: Piecewise linear regression — 1/Weight vs Nitrogen content (breakpoint at 1/W = 71.8, Adj R^2 = 0.545).

The breakpoint at 1/W ≈ 71.8 (corresponding to W ≈ 0.014 g) marks the boundary between the adult and larval weight ranges. Below the breakpoint (heavier specimens, small 1/W), N is relatively flat; above it (very light adults, large 1/W), N rises sharply. This two-slope structure is the formal statistical expression of the biological transition between developmental stages. Model M4 achieves Adj R^2 = 0.545, a substantial improvement over the simple linear models (M1: 0.08, M2: 0.32).

Figure 5 — GLM Fitted Lines by Stage × Location

The best-fitting model (M5: GLM with 1/Weight + Stage + Location) is visualised with four fitted lines — one per Stage × Location combination. Stage is encoded by colour, Location by line type and point shape.

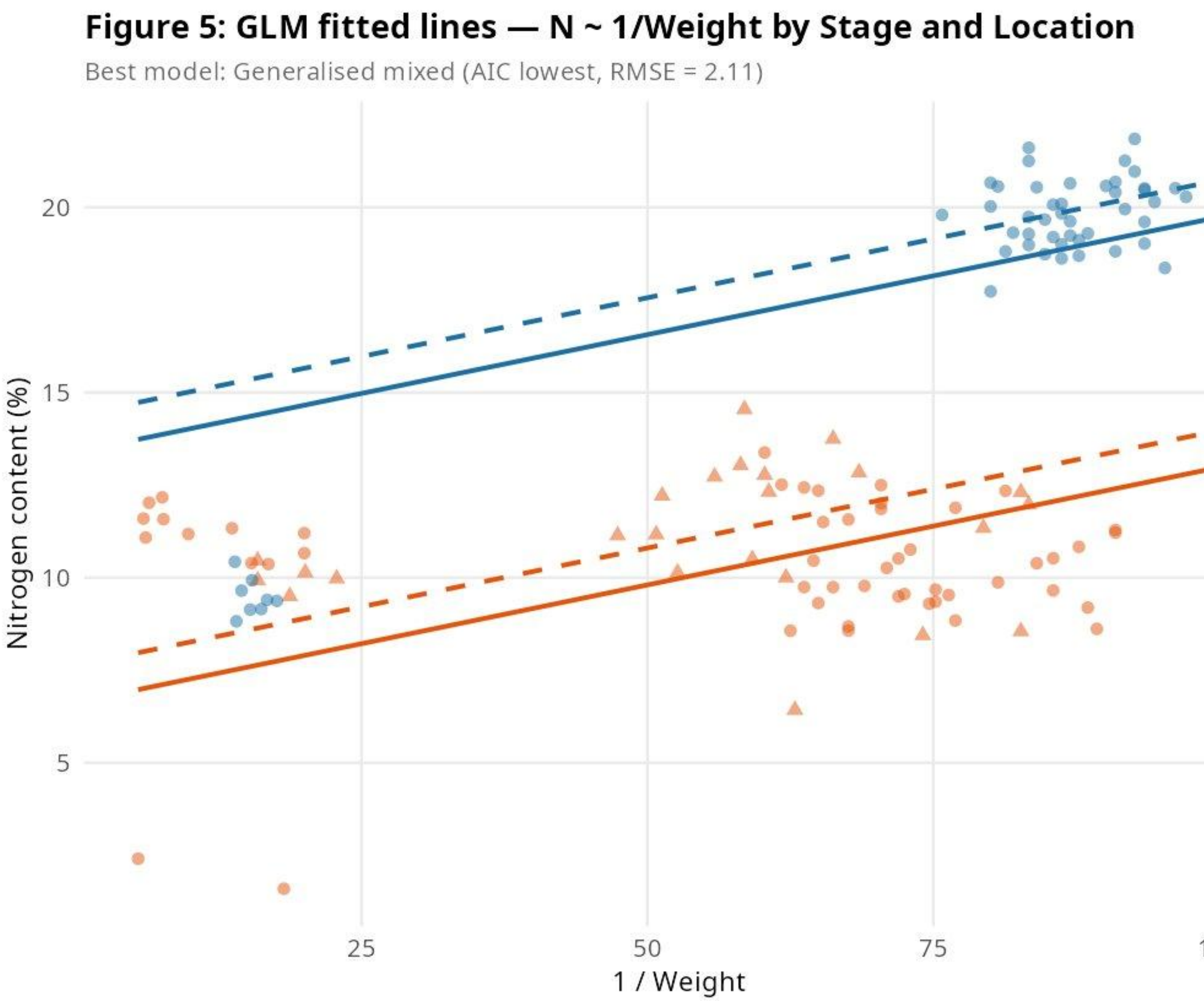

Figure 5: GLM fitted lines — N ~ 1/Weight by Stage and Location (best model, AIC lowest, RMSE = 2.11).

The separation between adult and larval fitted lines is dramatic, confirming Stage as the dominant predictor. The L1 vs L2 offset is smaller but consistent across both stages, capturing the location effect identified in Section 4a. The GLM outperforms all single-predictor models: including Stage and Location captures information about N that Weight alone cannot provide, confirming that the apparent weight–N relationship is largely a stage-composition artefact.

Model Comparison Summary

Model	Adj R^2	RMSE	AIC	Notes
M1: Linear (Weight)	0.08	~4.5	Highest	Underfits
M2: Linear (1/Weight)	0.32	~3.9	Lower	Reciprocal helps
M3: Piecewise (Weight)	0.12	~4.3	—	Modest gain over M1
M4: Piecewise (1/Weight)	0.545	~3.5	Lower	Best linear-family
M5: GLM (1/W+Stage+Loc)	—	~2.1	Lowest	✓ Best overall
M6: Stepwise GLM	—	~2.1	Near-low	Same predictors as M5

Reproduction Instructions

1. Clone the repository from github.com/KonGianniou 2. Run generate_synthetic_data_v2.R to produce Data_TP_synthetic.csv 3. Run analysis_v2.R — all packages install automatically on first run 4. Five figures are saved to the working directory at 150 dpi

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